数学笔记

乐绎华

2025 年 8 月 28 日

目录

1 Holomorphic forms and vector fields **6**

2 Hermitian bundles **9**

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Examples:

1.1. Preliminaries. Kähler manifolds may be considered as special Riemannian manifolds. Besides the Riemannian structure, they also have compatible symplectic and complex structures. Here are a few examples of Kähler manifolds:

- $(\mathbb{C}^m, \langle\ ,\ \rangle)$, where $\langle\ ,\ \rangle$ denotes the Hermitian metric $ds^2 = \mathrm{Re}(\sum dz_i d\bar{z}_i)$;
- any oriented 2–dimensional Riemannian manifold;
- the complex projective space $(\mathbb{C}\mathrm{P}^m, FS)$ endowed with the Fubini-Study metric;
- every projective manifold, that is, submanifold of $\mathbb{C}\mathrm{P}^m$ defined by homogeneous polynomials in $\mathbb{C}^{m+1}$.

Applications:

Kähler manifolds have found many applications in various domains like Differential Geometry, Complex Analysis, Algebraic Geometry or Theoretical Physics. To illustrate their importance let us make the following remark. With two exceptions (the so–called Joyce manifolds in dimensions 7 and 8), the only known compact examples of manifolds satisfying Einstein's equations

$$R_{\alpha\beta} = 0$$

(Ricci–flat in modern language) are constructed on Kähler manifolds. Generic Ricci–flat Kähler manifolds, also called *Calabi–Yau manifolds*, will be studied later on in these notes.

Definition 1.1. *A Kähler structure on a Riemannian manifold (M^n, g) is given by a 2–form Ω and a field of endomorphisms of the tangent bundle J satisfying the following*

- *algebraic conditions*
  a) *J is an almost complex structure: $J^2 = -Id$.*
  b) $g(X,Y) = g(JX, JY) \quad \forall\ X, Y \in TM$.
  c) $\Omega(X,Y) = g(JX, Y)$.
- *analytic conditions*
  d) *the 2–form Ω is symplectic: $d\Omega = 0$.*
  e) *J is integrable in the sense that its Nijenhuis tensor vanishes (see (4) below).*

Condition a) requires the real dimension of M to be even. Obviously, given the metric and one of the tensors J and Ω, we can immediately recover the other one by the formula c).

The motivation of the definition of J is from j, the multiplication by i on $\mathbb{C}$ via the identification of $\mathbb{R}^2$ with $\mathbb{C}$ given by $z = x + iy \mapsto (x, y)$. The endomorphism j can be expressed in the canonical base as

$$j = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

The differential of $F = f + ig : U \subset \mathbb{C} \to \mathbb{C}$ (viewed as a real function $F : U \subset \mathbb{R}^2 \to \mathbb{R}^2$) is the linear map

$$F_* = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}$$

Then it is easy to check that the Cauchy-Riemann relations are equivalent to the commutation relation $jF_* = F_* j$.

Then for $F : U \subset \mathbb{C}^n \to \mathbb{C}^m$, it is holomorphic iff the diffrerential F_* of F as real map $F : \mathbb{R}^{2n} \to \mathbb{R}^{2m}$ satisfies $j_m F_* = F_* j_n$, where

$$j_m = \begin{pmatrix} 0 & -I_m \\ I_m & 0 \end{pmatrix}$$

1.3. Complex manifolds. A complex manifold of complex dimension m is a topological space M with an open covering $\mathcal{U}$ such that for every point $x \in M$ there exists $U \in \mathcal{U}$ containing x and a homeomorphism $\phi_U : U \to \tilde{U} \subset \mathbb{C}^m$, such that for every intersecting $U, V \in \mathcal{U}$, the map between open sets of $\mathbb{C}^m$

$$\phi_{UV} := \phi_U \circ \phi_V^{-1}$$

is holomorphic. A pair (U, ϕ_U) is called a chart and the collection of all charts is called a *holomorphic structure.*

Important example. The complex projective space $\mathbb{C}\mathrm{P}^m$ can be defined as the set of complex lines of $\mathbb{C}^{m+1}$ (a line is a vector subspace of dimension one). If we define the equivalence relation $\sim$ on $\mathbb{C}^{m+1} - \{0\}$ by

$$(z_0, \ldots, z_m) \sim (\alpha z_0, \ldots, \alpha z_m), \qquad \forall\, \alpha \in \mathbb{C}^*,$$

then $\mathbb{C}\mathrm{P}^m = \mathbb{C}^{m+1} - \{0\}/\sim$. The equivalence class of $(z_0, \ldots z_m)$ will be denoted by $[z_0 : \ldots : z_m]$. Consider the open cover U_i, $i = 0, \ldots, m$ of $\mathbb{C}\mathrm{P}^m$ defined by

$$U_i := \{[z_0 : \ldots : z_m] \mid z_i \neq 0\}$$

and the maps $\phi_i : U_i \to \mathbb{C}^m$,

$$\phi_i([z_0 : \ldots : z_m]) = \left(\frac{z_0}{z_i}, \ldots, \frac{z_{i-1}}{z_i}, \frac{z_{i+1}}{z_i} \ldots, \frac{z_m}{z_i}\right).$$

A function $F : M \to \mathbb{C}$ is called holomorphic if $F \circ \phi_U^{-1}$ is holomorphic for every $U \in \mathcal{U}$. This property is local. To check it in the neighborhood of a point x it is enough to check it for a single $U \in \mathcal{U}$ containing x.

Almost complex structure J :

The most important object on a complex manifold from the differential geometric point of view is the almost complex structure J, which is a field of endomorphisms of the tangent bundle defined as follows. For every $X \in T_xM$, choose $U \in \mathcal{U}$ containing x and define

$$J_U(X) = (\phi_U)_*^{-1} \circ j_n \circ (\phi_U)_*(X).$$

J_U does not depend on U and their collection is a well-defined tensor J on M. This tensor clearly satisfies $J^2 = -id$.

Definition 1.2. *A (1,1)–tensor J on a differential manifold M satisfying $J^2 = -Id$ is called an almost complex structure. The pair (M, J) is then referred to as almost complex manifold.*

A $(1,1)$ -tensor J is a type of tensor that takes in one vector and one covector (a linear function) and outputs a scalar.

A complex manifold is thus in a canonical way an almost complex manifold.

1.4. The complexified tangent bundle. Let (M, J) be an almost complex manifold. We would like to diagonalize the endomorphism J. In order to do so, we have to complexify the tangent space. Define

$$TM^{\mathbb{C}} := TM \otimes_{\mathbb{R}} \mathbb{C}.$$

We extend all real endomorphisms and differential operators from TM to $TM^{\mathbb{C}}$ by $\mathbb{C}$–linearity. Let $T^{1,0}M$ (resp. $T^{0,1}M$) denote the eigenbundle of $TM^{\mathbb{C}}$ corresponding to the eigenvalue i (resp. $-i$) of J. The following algebraic lemma is an easy exercise.

LEMMA 1.3. *One has*

$$T^{1,0}M = \{X - iJX \mid X \in TM\}\ , \qquad T^{0,1}M = \{X + iJX \mid X \in TM\}.$$

and $TM^{\mathbb{C}} = T^{1,0}M \oplus T^{0,1}M$.

The famous Newlander–Nirenberg theorem can be stated as follows:

THEOREM 1.4. *Let (M, J) be an almost complex manifold. The almost complex structure J comes from a holomorphic structure if and only if the distribution $T^{0,1}M$ is integrable.*

Integrability of the distribution $T^{0,1}M$ means that its sections are closed under the Lie bracket, which is a condition that ensires the existence of a compatible complex coordinate system. In other words:

$$[X, Y] \in \Gamma(D) \qquad \forall X, Y \in \Gamma(D)$$

where the distribution D is a subbundle of $T_{\mathbb{C}}M$ and $\Gamma(D)$ is a vector field in D.

Suppose that J comes from a holomorphic structure on M. Consider a local chart (U, ϕ_U) and let $z_\alpha = x_\alpha + iy_\alpha$ be the α -th component of ϕ_U. If $\{e_1, \ldots, e_{2m}\}$ denotes the standard basis of $\mathbb{R}^{2m}$, we have by definition:

$$\frac{\partial}{\partial x_\alpha} = (\phi_U)_*^{-1}(e_\alpha), \qquad \frac{\partial}{\partial y_\alpha} = (\phi_U)_*^{-1}(e_{m+\alpha})$$

Moreover, $j_m(e_\alpha) = e_{m+\alpha}$, so we obtain directly from the defintion

$$J\left(\frac{\partial}{\partial x_\alpha}\right) = \frac{\partial}{\partial y_\alpha}$$

We now make the following notations

$$\frac{\partial}{\partial z_\alpha} := \frac{1}{2}\left(\frac{\partial}{\partial x_\alpha} - i\frac{\partial}{\partial y_\alpha}\right), \quad \frac{\partial}{\partial \bar{z}_\alpha} := \frac{1}{2}\left(\frac{\partial}{\partial x_\alpha} + i\frac{\partial}{\partial y_\alpha}\right).$$

Suppose that u_α, v_α is another local system of coordinates, satisfying

$$\frac{\partial}{\partial v_\alpha} = J\frac{\partial}{\partial u_\alpha}$$

We then have

$$\frac{\partial u_\beta}{\partial x_\alpha} = \frac{\partial v_\beta}{\partial y_\alpha} \quad \text{and} \quad \frac{\partial u_\beta}{\partial y_\alpha} = -\frac{\partial v_\beta}{\partial x_\alpha},$$

thus the transition functions are holomorphic.

(2) Let $A + iB \in \mathrm{Gl}_m(\mathbb{C})$. Compute the product

$$\begin{pmatrix} I_m & 0 \\ -iI_m & I_m \end{pmatrix}\begin{pmatrix} A & B \\ -B & A \end{pmatrix}\begin{pmatrix} I_m & 0 \\ iI_m & I_m \end{pmatrix}$$

and use this computation to prove that for every invertible matrix $A + iB \in \mathrm{Gl}_m(\mathbb{C})$, the determinant of the real $2m \times 2m$ matrix

$$\begin{pmatrix} A & B \\ -B & A \end{pmatrix}$$

is strictly positive.

练习 0.1.

证明.

$$\cdots = \begin{pmatrix} A + iB & B \\ O & A - iB \end{pmatrix}$$

Then

$$\det\begin{pmatrix} A & B \\ -B & A \end{pmatrix} = \det(A+iB)\cdot\det(A-iB) = \det(A+iB)\cdot\overline{\det(A+iB)} = |\det(A+iB)|^2 > 0$$

□

(3) Show that every almost complex manifold is orientable.

练习 0.2.

A smooth manifold is **orientable** if you can find an **oriented atlas**.

Given a collection of coordinate charts (U_k, ϕ_k) that covers the manifold, we need to show that the Jacobian determinant of all transition functions $\phi_j \circ \phi_k^{-1}$ is everywhere positive.

Under the basis $(v_1, \ldots, v_n, Jv_1, \ldots, Jv_n)$, any transition functions have the real matrix form

$$\begin{pmatrix} A & B \\ -B & A \end{pmatrix}$$

which has strictly positive determinant by theorem 0.1 .

1 Holomorphic forms and vector fields

2.1. Decomposition of the (complexified) exterior bundle. Let (M, J) be an almost complex manifold. We now turn our attention to exterior forms and introduce the complexified exterior bundle $\Lambda^*_{\mathbb{C}}M = \Lambda^*M \otimes_{\mathbb{R}} \mathbb{C}$. The sections of $\Lambda^*_{\mathbb{C}}M$ can be viewed as complex–valued forms or as formal sums $\omega + i\tau$, where ω and τ are usual real forms on M.

We define the following two sub–bundles of $\Lambda^1_{\mathbb{C}}M$:

$$\Lambda^{1,0}M := \{\xi \in \Lambda^1_{\mathbb{C}}M \mid \xi(Z) = 0 \ \forall \ Z \in T^{0,1}M\}$$

and

$$\Lambda^{0,1}M := \{\xi \in \Lambda^1_{\mathbb{C}}M \mid \xi(Z) = 0 \ \forall \ Z \in T^{1,0}M\}.$$

The sections of these sub–bundles are called forms of type (1,0) or forms of type (0,1) respectively. Similarly to Lemma 1.3 we have

Lemma 2.1. *One has*

$$\Lambda^{1,0}M = \{\omega - i\omega \circ J \mid \omega \in \Lambda^1 M\} \ , \qquad \Lambda^{0,1}M = \{\omega + i\omega \circ J \mid \omega \in \Lambda^1 M\}.$$

and $\Lambda^1_{\mathbb{C}}M = \Lambda^{1,0}M \oplus \Lambda^{0,1}M$.

Let us denote the k–th exterior power of $\Lambda^{1,0}$ (resp. $\Lambda^{0,1}$) by $\Lambda^{k,0}$ (resp. $\Lambda^{0,k}$) and let $\Lambda^{p,q}$ denote the tensor product $\Lambda^{p,0}\otimes\Lambda^{0,q}$. The exterior power of a direct sum of vector spaces can be described as follows

$$\Lambda^k(E\oplus F)\simeq\oplus_{i=0}^k\Lambda^iE\otimes\Lambda^{k-i}F.$$

Using Lemma 2.1 we then get

$$\Lambda^k_{\mathbb{C}}M\simeq\oplus_{p+q=k}\Lambda^{p,q}M.$$

Sections of $\Lambda^{p,q}M$ are called forms of type (p,q). It is easy to check that a complex–valued k–form ω is a section of $\Lambda^{k,0}M$ if and only if $Z\lrcorner\omega=0$ for all $Z\in T^{0,1}M$. More generally, a k–form is a section of $\Lambda^{p,q}M$ if and only if it vanishes whenever applied to $p+1$ vectors from $T^{1,0}M$ or to $q+1$ vectors from $T^{0,1}M$.

To every almost complex structure J one can associate a (2,1)–tensor N^J called the Nijenhuis tensor, satisfying

$$N^J(X,Y)=[X,Y]+J[JX,Y]+J[X,JY]-[JX,JY],\qquad\forall\ X,Y\in\mathcal{C}^\infty(TM). \tag{4}$$

$[\cdot,\cdot]$ is the Lie Bracket, which is defined by

$$[X,Y](f)=X(Y(f))-Y(X(f))$$

Proposition 2.2. *Let J be an almost complex structure on M^{2m}. The following statements are equivalent:*

(a) *J is a complex structure.*
(b) *$T^{0,1}M$ is integrable.*
(c) *$d\mathcal{C}^\infty(\Lambda^{1,0}M)\subset\mathcal{C}^\infty(\Lambda^{2,0}M\oplus\Lambda^{1,1}M)$.*
(d) *$d\mathcal{C}^\infty(\Lambda^{p,q}M)\subset\mathcal{C}^\infty(\Lambda^{p+1,q}M\oplus\Lambda^{p,q+1}M)\ \forall\ 0\le p,q\le m$.*
(e) *$N^J=0$.*

命题 1.1.

证明. Read. □

LEMMA 2.3. *Let $f : M \to \mathbb{C}$ be a smooth complex–valued function on M. The following assertions are equivalent:*

(1) *f is holomorphic.*
(2) $Z(f) = 0 \ \ \forall\ Z \in T^{0,1}M$.
(3) *df is a form of type $(1,0)$.*

PROOF. (2) $\iff$ (3). $df \in \Lambda^{1,0}M \iff df(Z) = 0\ \forall\ Z \in T^{0,1}M \iff Z(f) = 0 \ \ \forall\ Z \in T^{0,1}M$.

(1) $\iff$ (3). The function f is holomorphic if and only if $f \circ \phi_U^{-1}$ is holomorphic for every holomorphic chart (U, ϕ_U), which is equivalent to $f_* \circ (\phi_U)_*^{-1} \circ j_m = if_* \circ (\phi_U)_*^{-1}$, that is, $f_* \circ J = if_*$. This last equation just means that for every real vector X, $df(JX) = idf(X)$, hence $idf(X + iJX) = 0 \ \ \forall\ X \in TM$, which is equivalent to $df \in \Lambda^{1,0}M$. □

Using Proposition 2.2, for every fixed (p, q) we define the differential operators $\partial : \mathcal{C}^\infty(\Lambda^{p,q}M) \to \mathcal{C}^\infty(\Lambda^{p+1,q}M)$ and $\bar\partial : \mathcal{C}^\infty(\Lambda^{p,q}M) \to \mathcal{C}^\infty(\Lambda^{p,q+1}M)$ by $d = \partial + \bar\partial$.

LEMMA 2.4. *The following identities hold:*

$$\partial^2 = 0, \qquad \bar\partial^2 = 0, \qquad \partial\bar\partial + \bar\partial\partial = 0.$$

PROOF. We have $0 = d^2 = (\partial + \bar\partial)^2 = \partial^2 + \bar\partial^2 + (\partial\bar\partial + \bar\partial\partial)$, and the three operators in the last term take values in different sub–bundles. □

DEFINITION 2.5. *A vector field Z in $\mathcal{C}^\infty(T^{1,0}M)$ is called holomorphic if $Z(f)$ is holomorphic for every locally defined holomorphic function f. A p–form ω of type $(p, 0)$ is called holomorphic if $\bar\partial\omega = 0$.*

DEFINITION 2.6. *A real vector field X is called real holomorphic if $X - iJX$ is a holomorphic vector field.*

LEMMA 2.7. *Let X be a real vector field on a complex manifold (M, J). The following assertions are equivalent:*

- *X is real holomorphic*
- $\mathcal{L}_X J = 0$
- *The flow of X consists of holomorphic transformations of M.*

Although not explicitly stated, the reader might have guessed that a map $f : (M, J_1) \to (N, J_2)$ between two complex manifolds is called holomorphic if its differential commutes with the complex structures at each point: $f_* \circ J_1 = J_2 \circ f_*$.

We close this section with the following important result:

PROPOSITION 2.8. (The local $i\partial\bar\partial$–Lemma). *Let $\omega \in \Lambda^{1,1}M \cap \Lambda^2 M$ be a real 2–form of type (1,1) on a complex manifold M. Then ω is closed if and only if every point $x \in M$ has an open neighborhood U such that $\omega|_U = i\partial\bar\partial u$ for some real function u on U.*

PROOF. One implication is clear from Lemma 2.4:

$$d(i\partial\bar\partial) = i(\partial + \bar\partial)\partial\bar\partial = i(\partial^2\bar\partial - \partial\bar\partial^2) = 0.$$

The other implication is more delicate and needs the following counterpart of the Poincaré Lemma (see [**2**] p. 25 for a proof):

LEMMA 2.9. $\bar{\partial}$–Poincaré Lemma. *A $\bar{\partial}$–closed (0,1)–form is locally $\bar{\partial}$–exact.*

Let ω be a closed real form of type (1,1). From the Poincaré Lemma, there exists locally a 1–form τ with $d\tau = \omega$. Let $\tau = \tau^{1,0} + \tau^{0,1}$ be the decomposition of τ in forms of type (1,0) and (0,1). Clearly, $\tau^{1,0} = \overline{\tau^{0,1}}$. Comparing types in the equality

$$\omega = d\tau = \bar{\partial}\tau^{0,1} + (\partial\tau^{0,1} + \bar{\partial}\tau^{1,0}) + \partial\tau^{1,0},$$

we get $\bar{\partial}\tau^{0,1} = 0$ and $\omega = (\partial\tau^{0,1}+\bar{\partial}\tau^{1,0})$. The $\bar{\partial}$–Poincaré Lemma yields a local function f such that $\tau^{0,1} = \bar{\partial}f$. By conjugation we get $\tau^{1,0} = \partial\bar{f}$, hence $\omega = (\partial\tau^{0,1}+\bar{\partial}\tau^{1,0}) = \partial\bar{\partial}f+\bar{\partial}\partial\bar{f} = i\partial\bar{\partial}(2\mathrm{Im}(f))$, and the Proposition follows, with $u := 2\mathrm{Im}(f)$. □

2 Hermitian bundles

Let M be a differentiable manifold (not necessarily complex) and let $E \to M$ be a complex vector bundle over M, which means E is a space with a projection map to M, and the fibers over each point in M are complex vector space.

定义 2.1 (connection)**.** A ($\mathbb{C}$ -linear) connection on a E is a $\mathbb{C}$ -linear differential operator $\nabla : C^\infty(E) \to C^\infty(\Lambda^1(E))$ satisfying the Leibniz rule

$$\nabla(f\sigma) = df \otimes \sigma + f\nabla\sigma, \qquad \forall f \in C^\infty(M)$$

$C^\infty(\Lambda^1(E))$ is the space of E -valued 1-forms, where $\Lambda^1(E)$ denotes the bundle of E -valued 1-forms over M.

One can extend any connection to the bundles of E -valued p -forms on M by

$$\nabla(\omega \otimes \sigma) = d\omega \otimes \sigma + (-1)^p \omega \wedge \nabla\sigma$$

where the wedge product has to be understood as

$$\omega \wedge \nabla\sigma := \sum_{i=1}^{n} \omega \wedge e_i^* \otimes \nabla_{e_i}\sigma$$

for any local basis $\{e_i\}$ of TM with dual basis $\{e_i^*\}$. $\nabla_{e_i}\sigma$ is the covariant derivative along the basis vector e_i.

定义 2.2 (curvature)**.** The **curvature** of ∇ is the $\mathrm{End}(E)$ -valued 2-form R^∇ defined by

$$R^\nabla(\sigma) \coloneqq \nabla(\nabla\sigma) \qquad \forall \sigma \in C^\infty(E)$$

To check that this is indeed tensorial (i.e. depends on σ and not on f), we can write

$$\nabla^2(f\sigma) = \nabla(df \otimes \sigma + f\nabla\sigma) = \underbrace{d^2 f}_{=0} \otimes \sigma - df \wedge \nabla\sigma + df \wedge \nabla\sigma + f\nabla^2\sigma = f\nabla^2\sigma$$

More explicitly, if $\{\sigma_1, \dots, \sigma_k\}$ are local sections of E which form a basis of each fiber over some open set U, we define the connection forms $\omega_{ij} \in \Lambda^1(U)$ (relative to the choice of the base) by

$$\nabla\sigma_i = \omega_{ij} \otimes \sigma_j.$$

We define the two–forms R^∇_{ij} by

$$R^\nabla(\sigma_i) = R^\nabla_{ij} \otimes \sigma_j,$$

and compute

$$R^\nabla_{ij} \otimes \sigma_j = R^\nabla(\sigma_i) = \nabla(\omega_{ij} \otimes \sigma_j) = (d\omega_{ij}) \otimes \sigma_j - \omega_{ik} \wedge \omega_{kj} \otimes \sigma_j,$$

showing that

$$R^\nabla_{ij} = d\omega_{ij} - \omega_{ik} \wedge \omega_{kj}. \tag{9}$$

4.2. Hermitian structures and connections. Let $E \to M$ be a complex rank k bundle over some manifold M. We do not assume for the moment that M has an almost complex structure.

Definition 4.2. *A* Hermitian structure *H on E is a smooth field of Hermitian products on the fibers of E, that is, for every $x \in M$, $H : E_x \times E_x \to \mathbb{C}$ satisfies*

- *$H(u, v)$ is $\mathbb{C}$–linear in u for every $v \in E_x$.*
- *$H(u, v) = \overline{H(v, u)} \quad \forall\ u, v \in E_x$.*

- *$H(u, u) > 0 \quad \forall\ u \neq 0$.*
- *$H(u, v)$ is a smooth function on M for every smooth sections u, v of E.*